

Revisiting a random model of lateral gene transfer in phylogenetics

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This Mathematica notebook accompanies the manuscript and provides the full set of calculations underlying the results presented in the main text.

The 2-taxon case

Species tree: $T = (1, 2)$

Arc notation: for $i = 1, 2$, e_i denotes the arc incident to leaf i .

Furthermore, t_1 denotes the t-value of the interior vertex of T , h the t-value of the relevant transfer event, and g the t-value of the interior vertex of the resulting gene histories.

Gene history densities

H0: No transfer event; $g=t_1$

```
In[*]:= fH0 = Exp[-2 * λ * t1]
Out[*]:= e-2 t1 λ
```

H1: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_1 to e_2 or from e_2 to e_1 ; $g=h$; $0 < h < t_1$.

```
In[*]:= fH1 = 2 * 1 / 2 * λ * Exp[-2 * λ * h]
Out[*]:= 2 e-2 h λ λ
```

Gene tree probabilities

```
In[*]:= ProbT = fH0 + Integrate[fH1, {h, 0, t1}]
Out[*]:= 1
```

Site pattern probabilities under the JC69 model

Transition probabilities under JC for tree (1:g, 2:g) derived analogously to small phylogenetic trees website (https://www.coloradocollege.edu/aapps/ldg/small-trees/small-trees_0.html).

```
In[*]:= ProbXX[g_, μ_] := Module[{b0, b1, pxx},
  b0 = 1/4 + 3/4 * Exp[-g * μ];
  b1 = 1/4 - 1/4 * Exp[-g * μ];

  pxx = b0^2 + 3 * b1^2
];

In[*]:= ProbXY[g_, μ_] := Module[{b0, b1, pxy},
  b0 = 1/4 + 3/4 * Exp[-g * μ];
  b1 = 1/4 - 1/4 * Exp[-g * μ];

  pxy = 6 * b0 * b1 + 6 * b1^2
];

In[*]:= ExpandAll[ProbXX[1, 4/3] + ProbXY[1, 4/3]]
Out[*]=
1
```

Pxx

```
In[*]:= Pxx = FullSimplify[ExpandAll[fH0 * ProbXX[t1, 4/3]] +
  ExpandAll[Integrate[fH1 * ProbXX[h, 4/3], {h, 0, t1}]]]
Out[*]=

$$\frac{1 + 3 e^{-\frac{2}{3} t_1 (4+3 \lambda)} + 3 \lambda}{4 + 3 \lambda}$$

```

Pxy

```
In[*]:= Pxy = FullSimplify[ExpandAll[fH0 * ProbXY[t1, 4/3]] +
  ExpandAll[Integrate[fH1 * ProbXY[h, 4/3], {h, 0, t1}]]]
Out[*]=

$$\frac{3 - 3 e^{-\frac{2}{3} t_1 (4+3 \lambda)}}{4 + 3 \lambda}$$

```

The 3-taxon case

Species tree: $S = ((1, 2), 3)$.

Arc notation: for $(i = 1, 2, 3)$, let e_i denote the arc incident to leaf i ; let e_4 denote the interior arc.

Moreover, let $t_1 < t_2$ be the t -values of the interior vertices of S ; let $h_1 < h_2$ be the t -values of the relevant transfer events; and let $g_1 < g_2$ be the t -values of the interior vertices of the resulting gene histories.

Gene history densities

G1x: No transfer event; $g_1=t_1, g_2=t_2$

$$\ln[*] := f_{G1x} = \text{Exp}[-\lambda * (t_1 + 2 * t_2)];$$

G1a: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_1 to e_2 ; $g_1=h_1, g_2=t_2$; $0 < h_1 < t_1$

$$\ln[*] := f_{G1a} = (1 / 2) * \lambda * \text{Exp}[-\lambda * (2 * t_2 + h_1)];$$

G1b: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_2 to e_1 ; $g_1=h_1, g_2=t_2$; $0 < h_1 < t_1$

$$\ln[*] := f_{G1b} = (1 / 2) * \lambda * \text{Exp}[-\lambda * (2 * t_2 + h_1)];$$

G1c: First relevant transfer in $(0, t_1)$ with transfer arc between e_1 and e_2 (direction irrelevant); second relevant transfer also in $(0, t_1)$ with transfer arc between e_1 and e_3 (direction irrelevant); $g_1=h_1, g_2=h_2$; $0 < h_1 < t_1$ and $0 < h_2 < t_1 - h_1$

$$\ln[*] := f_{G1c} = 2 * \lambda^2 * \text{Exp}[-\lambda * (3 * h_1 + 2 * h_2)];$$

G1d: First relevant transfer in $(0, t_1)$ with transfer arc between e_1 and e_2 (direction irrelevant); second relevant transfer in (t_1, t_2) with transfer arc between e_3 and e_4 (direction irrelevant); $g_1=h_1, g_2=h_2$; $0 < h_1 < t_1$ and $0 < h_2 < t_2 - t_1$

$$\ln[*] := f_{G1d} = 2 * \lambda^2 * \text{Exp}[-\lambda * (2 * t_1 + h_1 + 2 * h_2)];$$

G1y: First relevant transfer in (t_1, t_2) with transfer arc between e_3 and e_4 (direction does not matter); $g_1=t_1, g_2=h_1, 0 < h_1 < t_2 - t_1$

$$\ln[*] := f_{G1y} = 2 * \lambda * \text{Exp}[-\lambda * (3 * t_1 + 2 * h_1)];$$

G2a: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_1 to e_3 ; $g_1 = h_1, g_2=t_1$; $0 < h_1 < t_1$

$$\ln[*] := f_{G2a} = (1 / 2) * \lambda * \text{Exp}[-\lambda * (2 * t_1 + h_1)];$$

G2b: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_3 to e_1 ; $g_1=h_1, g_2 = t_2$; $0 < h_1 < t_1$

$$\ln[*] := f_{G2b} = (1 / 2) * \lambda * \text{Exp}[-\lambda * (2 * t_2 + h_1)];$$

G2c: First relevant transfer in $(0, t_1)$ with transfer arc between e_1 and e_3 (direction irrelevant) and

second relevant transfer in $(0, t_1)$ between e_1 and e_2 (direction irrelevant); $g_1 = h_1, g_2 = h_2, 0 < h_1 < t_1$ and $0 < h_2 < t_1 - h_1$

```
In[*]:= fG2c = 2 * λ^2 * Exp[-λ * (3 * h1 + 2 * h2)] ;
```

G2d: First relevant transfer in $(0, t_1)$ with transfer arc from e_3 to e_1 and second relevant transfer in (t_1, t_2) between e_3 and e_4 (direction irrelevant); $g_1 = h_1, g_2 = h_2; 0 < h_1 < t_1$ and $0 < h_2 < t_2 - t_1$

```
In[*]:= fG2d = λ^2 * Exp[-λ * (2 * t1 + h1 + 2 * h2)] ;
```

G3a: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_2 to e_3 ; $g_1 = h_1, g_2 = t_1; 0 < h_1 < t_1$

```
In[*]:= fG3a = (1 / 2) * λ * Exp[-λ * (2 * t1 + h1)] ;
```

G3b: Precisely one relevant transfer in $(0, t_1)$ with transfer arc from e_3 to e_2 ; $g_1 = h_1, g_2 = t_2; 0 < h_1 < t_1$

```
In[*]:= fG3b = (1 / 2) * λ * Exp[-λ * (2 * t2 + h1)] ;
```

G3c: First relevant transfer in $(0, t_1)$ with transfer arc between e_1 and e_2 (direction irrelevant) and second relevant transfer in $(0, t_1)$ between e_1 and e_2 (direction irrelevant); $g_1 = h_1, g_2 = h_2, 0 < h_1 < t_1$ and $0 < h_2 < t_1 - h_1$

```
In[*]:= fG3c = 2 * λ^2 * Exp[-λ * (3 * h1 + 2 * h2)] ;
```

G3d: First relevant transfer in $(0, t_1)$ with transfer arc from e_3 to e_2 and second relevant transfer in (t_1, t_2) between e_3 and e_4 (direction irrelevant); $g_1 = h_1, g_2 = h_2; 0 < h_1 < t_1$ and $0 < h_2 < t_2 - t_1$

```
In[*]:= fG3d = λ^2 * Exp[-λ * (2 * t1 + h1 + 2 * h2)] ;
```

Gene tree probabilities

$T_1 = ((1, 2), 3)$

```
In[*]:= ProbT1 = ExpandAll[
  TrigToExp[fG1x + Integrate[fG1a, {h1, 0, t1}] + Integrate[fG1b, {h1, 0, t1}] +
    Integrate[Integrate[fG1c, {h2, 0, t1 - h1}], {h1, 0, t1}] +
    Integrate[Integrate[fG1d, {h2, 0, t2 - t1}], {h1, 0, t1}] +
    Integrate[fG1y, {h1, 0, t2 - t1}]]]
```

```
Out[*]=
1   2
-- + -- e-3 t1 λ
3   3
```

$T_2 = ((1, 3), 2)$

```
In[*]:= ProbT2 =
  ExpandAll[TrigToExp[Integrate[fG2a, {h1, 0, t1}] + Integrate[fG2b, {h1, 0, t1}] +
    Integrate[Integrate[fG2c, {h2, 0, t1 - h1}], {h1, 0, t1}] +
    Integrate[Integrate[fG2d, {h2, 0, t2 - t1}], {h1, 0, t1}]]]
```

```
Out[*]=

$$\frac{1}{3} - \frac{1}{3} e^{-3 t_1 \lambda}$$

```

T3 = ((2,3),1)

```
In[*]:= ProbT3 =
  ExpandAll[TrigToExp[Integrate[fG3a, {h1, 0, t1}] + Integrate[fG3b, {h1, 0, t1}] +
    Integrate[Integrate[fG3c, {h2, 0, t1 - h1}], {h1, 0, t1}] +
    Integrate[Integrate[fG3d, {h2, 0, t2 - t1}], {h1, 0, t1}]]]
```

```
Out[*]=

$$\frac{1}{3} - \frac{1}{3} e^{-3 t_1 \lambda}$$

```

Site pattern probabilities under the JC69 model

Transition probabilities under JC for tree ((1:g1, 2:g1),3:g2) from the small phylogenetic trees website (https://www.coloradocollege.edu/aapps/ldg/small-trees/small-trees_0.html)

```
In[*]:= ProbXXX[g1_, g2_, μ_] := Module[{d0, d1, b0, b1, pxxx},
  d0 = 1/4 + 3/4 * Exp[-g1 * μ];
  d1 = 1/4 - 1/4 * Exp[-g1 * μ];
  b0 = 1/4 + 3/4 * Exp[-(g2 - g1) * μ];
  b1 = 1/4 - 1/4 * Exp[-(g2 - g1) * μ];

  pxxx = b0^2 * d0^3 + 3 * b0^2 * d1^3 +
    6 * b0 * b1 * d0^2 * d1 + 6 * b0 * b1 * d0 * d1^2 + 12 * b0 * b1 * d1^3 +
    3 * b1^2 * d0^3 + 6 * b1^2 * d0^2 * d1 + 6 * b1^2 * d0 * d1^2 + 21 * b1^2 * d1^3
];
```

```
In[*]:= ProbXXY[g1_, g2_, μ_] := Module[{d0, d1, b0, b1, pxy},
  d0 = 1/4 + 3/4 * Exp[-g1 * μ];
  d1 = 1/4 - 1/4 * Exp[-g1 * μ];
  b0 = 1/4 + 3/4 * Exp[-(g2 - g1) * μ];
  b1 = 1/4 - 1/4 * Exp[-(g2 - g1) * μ];

  pxy = 6 * b0^2 * d0^2 * d1 + 6 * b0^2 * d0 * d1^2 + 12 * b0^2 * d1^3 +
    12 * b0 * b1 * d0^2 * d1 + 84 * b0 * b1 * d0 * d1^2 + 48 * b0 * b1 * d1^3 +
    30 * b1^2 * d0^2 * d1 + 102 * b1^2 * d0 * d1^2 + 84 * b1^2 * d1^3
];
```

```

In[*]:= ProbXXY[g1_, g2_,  $\mu$ _] := Module[{d0, d1, b0, b1, pxy},
  d0 = 1/4 + 3/4 * Exp[-g1 *  $\mu$ ];
  d1 = 1/4 - 1/4 * Exp[-g1 *  $\mu$ ];
  b0 = 1/4 + 3/4 * Exp[-(g2 - g1) *  $\mu$ ];
  b1 = 1/4 - 1/4 * Exp[-(g2 - g1) *  $\mu$ ];

  pxy =
    1/2 * (3 * b0^2 * d0^2 * d1 + 3 * b0^2 * d0 * d1^2 + 6 * b0^2 * d1^3 + 6 * b0 * b1 * d0^3 +
      12 * b0 * b1 * d0^2 * d1 + 12 * b0 * b1 * d0 * d1^2 + 42 * b0 * b1 * d1^3 + 6 * b1^2 * d0^3 +
      21 * b1^2 * d0^2 * d1 + 21 * b1^2 * d0 * d1^2 + 60 * b1^2 * d1^3)
];

In[*]:= ProbXYZ[g1_, g2_,  $\mu$ _] := Module[{d0, d1, b0, b1, pxyz},
  d0 = 1/4 + 3/4 * Exp[-g1 *  $\mu$ ];
  d1 = 1/4 - 1/4 * Exp[-g1 *  $\mu$ ];
  b0 = 1/4 + 3/4 * Exp[-(g2 - g1) *  $\mu$ ];
  b1 = 1/4 - 1/4 * Exp[-(g2 - g1) *  $\mu$ ];

  pxyz =
    18 * b0^2 * d0 * d1^2 + 6 * b0^2 * d1^3 + 24 * b0 * b1 * d0^2 * d1 + 60 * b0 * b1 * d0 * d1^2 +
    60 * b0 * b1 * d1^3 + 24 * b1^2 * d0^2 * d1 + 114 * b1^2 * d0 * d1^2 + 78 * b1^2 * d1^3
];

In[*]:= FullSimplify[ProbXXX[1, 2.5, 4/3] +
  ProbXXY[1, 2.5, 4/3] + 2 * ProbXXY[1, 2.5, 4/3] + ProbXYZ[1, 2.5, 4/3]]
Out[*]=
1.

```

Pxxx

Use ProbXXX-function for all gene histories

```

Pxxx = FullSimplify[ExpandAll[fG1x*ProbXXX[t1, t2, 4/3]] +
  ExpandAll[Integrate[fG1a*ProbXXX[h1, t2, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[fG1b*ProbXXX[h1, t2, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[Integrate[fG1c*ProbXXX[h1, h2, 4/3], {h2, 0, t1 - h1}],
    {h1, 0, t1}]] + ExpandAll[Integrate[
    Integrate[fG1d*ProbXXX[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
  ExpandAll[Integrate[fG1y*ProbXXX[t1, h1, 4/3], {h1, 0, t2 - t1}]] +
  ExpandAll[Integrate[fG2a*ProbXXX[h1, t1, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[fG2b*ProbXXX[h1, t2, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[Integrate[fG2c*ProbXXX[h1, h2, 4/3], {h2, 0, t1 - h1}],
    {h1, 0, t1}]] + ExpandAll[Integrate[
    Integrate[fG2d*ProbXXX[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
  ExpandAll[Integrate[fG3a*ProbXXX[h1, t1, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[fG3b*ProbXXX[h1, t2, 4/3], {h1, 0, t1}]] +
  ExpandAll[Integrate[
    Integrate[fG3c*ProbXXX[h1, h2, 4/3], {h2, 0, t1 - h1}], {h1, 0, t1}]] +
  ExpandAll[
    Integrate[Integrate[fG3d*ProbXXX[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]]]

```

Out[8]=

$$\begin{aligned}
& \frac{(2+9\lambda)(4+3\lambda)(10+9\lambda)}{(4+3\lambda)(4+9\lambda)(8+9\lambda)} + \\
& \frac{3}{8} e^{-\frac{1}{3}(3t_1+2t_2)(4+3\lambda)} \left(-e^{2t_1(2+\lambda)} + \frac{3e^{\frac{2}{3}t_1(8+3\lambda)}\lambda(-4+3\lambda)}{(4+3\lambda)^2} - \frac{12e^{\frac{20}{3}t_1+3t_1\lambda}\lambda(2+3\lambda)}{(4+3\lambda)^2} + \right. \\
& \frac{12e^{\frac{8}{3}t_2+2t_2\lambda+t_1(4+\lambda)}\lambda(2+3\lambda)}{(4+3\lambda)^2} + \frac{e^{\frac{8}{3}t_1+2t_1\lambda}(4-3\lambda)}{4+3\lambda} + \frac{3e^{\frac{20}{3}t_1+2t_1\lambda}\lambda}{4+3\lambda} - \\
& \frac{3e^{\frac{8}{3}t_2+2t_2\lambda}\lambda}{4+3\lambda} + \frac{4e^{t_1(4+3\lambda)}(2+3\lambda)}{4+3\lambda} + \frac{3e^{4t_1+\frac{8}{3}t_2+2t_2\lambda}\lambda(16+3\lambda)}{(-8+3\lambda)(4+3\lambda)} - \\
& \frac{e^{\frac{4}{3}(t_1+2t_2)+2t_2\lambda}(4+9\lambda)}{8+9\lambda} - \frac{4e^{\frac{1}{3}(t_1+2t_2)(4+3\lambda)}(-32+3\lambda(-4+3\lambda)(-4+3\lambda))}{(-8+3\lambda)(-4+3\lambda)(4+3\lambda)} + \\
& \left. \frac{3e^{\frac{8}{3}(t_1+t_2)+2t_2\lambda}\lambda(-64+3\lambda(80+3\lambda(68+9\lambda)))}{(-4+3\lambda)(4+3\lambda)^2(4+9\lambda)} \right)
\end{aligned}$$

$ln[*] := \text{LGTProbXXX}[t1_ , t2_ , \lambda_] :=$

$$\begin{aligned} & \frac{(2+9\lambda)(4+3\lambda(10+9\lambda))}{(4+3\lambda)(4+9\lambda)(8+9\lambda)} + \frac{3}{8} e^{-\frac{1}{3}(3t1+2t2)(4+3\lambda)} \left(-e^{2t1(2+\lambda)} + \frac{3e^{\frac{2}{3}t1(8+3\lambda)}\lambda(-4+3\lambda)}{(4+3\lambda)^2} - \right. \\ & \frac{12e^{\frac{20t1}{3}+3t1\lambda}\lambda(2+3\lambda)}{(4+3\lambda)^2} + \frac{12e^{\frac{8t2}{3}+2t2\lambda+t1(4+\lambda)}\lambda(2+3\lambda)}{(4+3\lambda)^2} + \frac{e^{\frac{8t1}{3}+2t1\lambda}(4-3\lambda)}{4+3\lambda} + \\ & \frac{3e^{\frac{20t1}{3}+2t1\lambda}\lambda}{4+3\lambda} - \frac{3e^{\frac{8t2}{3}+2t2\lambda}\lambda}{4+3\lambda} + \frac{4e^{t1(4+3\lambda)}(2+3\lambda)}{4+3\lambda} + \frac{3e^{4t1+\frac{8t2}{3}+2t2\lambda}\lambda(16+3\lambda)}{(-8+3\lambda)(4+3\lambda)} - \\ & \frac{e^{\frac{4}{3}(t1+2t2)+2t2\lambda}(4+9\lambda)}{8+9\lambda} - \frac{4e^{\frac{1}{3}(t1+2t2)(4+3\lambda)}(-32+3\lambda(-4+3\lambda)(-4+3\lambda))}{(-8+3\lambda)(-4+3\lambda)(4+3\lambda)} + \\ & \left. \frac{3e^{\frac{8(t1+t2)}{3}+2t2\lambda}\lambda(-64+3\lambda(80+3\lambda(68+9\lambda)))}{(-4+3\lambda)(4+3\lambda)^2(4+9\lambda)} \right); \end{aligned}$$

Pxxy

Use ProbXXY-function for all gene histories $G_{\{1\}}$ matching the species tree and use ProbXYY-function for $G_{\{2\}}$ and $G_{\{3\}}$


```

In[*]:= Pxy = FullSimplify[ExpandAll[fG1x*ProbXXY[t1, t2, 4/3]] +
ExpandAll[Integrate[fG1a*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG1c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG1d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1y*ProbXXY[t1, h1, 4/3], {h1, 0, t2 - t1}]] +
ExpandAll[Integrate[fG2a*ProbXXY[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG2b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG2c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG2d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3a*ProbXXY[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[
Integrate[fG3c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}], {h1, 0, t1}]] +
ExpandAll[
Integrate[Integrate[fG3d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]]]

```

Out[*]=

$$\begin{aligned}
& \frac{12 (2 + \lambda (10 + 9 \lambda))}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \\
& \frac{3}{16} e^{-\frac{1}{3} (3 t_1 + 2 t_2) (4 + 3 \lambda)} \left(e^{2 t_1 (2 + \lambda)} + \frac{18 e^{\frac{20 t_1}{3} + 3 t_1 \lambda} (-2 + \lambda) \lambda}{(4 + 3 \lambda)^2} - \frac{18 e^{\frac{8 t_2}{3} + 2 t_2 \lambda + t_1 (4 + \lambda)} (-2 + \lambda) \lambda}{(4 + 3 \lambda)^2} - \right. \\
& \frac{3 e^{\frac{2}{3} t_1 (8 + 3 \lambda)} \lambda (-16 + 3 \lambda)}{(4 + 3 \lambda)^2} - \frac{6 e^{t_1 (4 + 3 \lambda)} (-2 + \lambda)}{4 + 3 \lambda} - \frac{3 e^{\frac{20 t_1}{3} + 2 t_1 \lambda} \lambda}{4 + 3 \lambda} + \\
& \frac{3 e^{\frac{8 t_2}{3} + 2 t_2 \lambda} \lambda}{4 + 3 \lambda} + \frac{e^{\frac{8 t_1}{3} + 2 t_1 \lambda} (-16 + 3 \lambda)}{4 + 3 \lambda} - \frac{4 e^{\frac{4}{3} (t_1 + 2 t_2) + 2 t_2 \lambda} (2 + 3 \lambda)}{8 + 9 \lambda} + \\
& \frac{2 e^{4 t_1 + \frac{8 t_2}{3} + 2 t_2 \lambda} (-16 + 5 \lambda (-2 + 3 \lambda))}{(-8 + 3 \lambda) (4 + 3 \lambda)} + \frac{2 e^{\frac{1}{3} (t_1 + 2 t_2) (4 + 3 \lambda)} (-64 + 3 \lambda (-8 - 30 \lambda + 9 \lambda^2))}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} - \\
& \left. \frac{3 e^{\frac{8 (t_1 + t_2)}{3} + 2 t_2 \lambda} \lambda (-256 + 3 \lambda (80 + 456 \lambda + 81 \lambda^2))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} \right)
\end{aligned}$$

$ln[*] := \text{LGTProbXXY}[t1_ , t2_ , \lambda_] :=$

$$\begin{aligned} & \frac{12 (2 + \lambda (10 + 9 \lambda))}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \frac{3}{16} e^{-\frac{1}{3} (3 t1 + 2 t2) (4 + 3 \lambda)} \left(e^{2 t1 (2 + \lambda)} + \frac{18 e^{\frac{20 t1}{3} + 3 t1 \lambda} (-2 + \lambda) \lambda}{(4 + 3 \lambda)^2} - \right. \\ & \frac{18 e^{\frac{8 t2}{3} + 2 t2 \lambda + t1 (4 + \lambda)} (-2 + \lambda) \lambda}{(4 + 3 \lambda)^2} - \frac{3 e^{\frac{2}{3} t1 (8 + 3 \lambda)} \lambda (-16 + 3 \lambda)}{(4 + 3 \lambda)^2} - \frac{6 e^{t1 (4 + 3 \lambda)} (-2 + \lambda)}{4 + 3 \lambda} - \\ & \frac{3 e^{\frac{20 t1}{3} + 2 t1 \lambda} \lambda}{4 + 3 \lambda} + \frac{3 e^{\frac{8 t2}{3} + 2 t2 \lambda} \lambda}{4 + 3 \lambda} + \frac{e^{\frac{8 t1}{3} + 2 t1 \lambda} (-16 + 3 \lambda)}{4 + 3 \lambda} - \frac{4 e^{\frac{4}{3} (t1 + 2 t2) + 2 t2 \lambda} (2 + 3 \lambda)}{8 + 9 \lambda} + \\ & \frac{2 e^{4 t1 + \frac{8 t2}{3} + 2 t2 \lambda} (-16 + 5 \lambda (-2 + 3 \lambda))}{(-8 + 3 \lambda) (4 + 3 \lambda)} + \frac{2 e^{\frac{1}{3} (t1 + 2 t2) (4 + 3 \lambda)} (-64 + 3 \lambda (-8 - 30 \lambda + 9 \lambda^2))}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} - \\ & \left. \frac{3 e^{\frac{8 (t1 + t2)}{3} + 2 t2 \lambda} \lambda (-256 + 3 \lambda (80 + 456 \lambda + 81 \lambda^2))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} \right); \end{aligned}$$

$P_{xyy} = P_{xyx}$

Use ProbXXY-function for $G_{\{1\}}$, use ProbXXY-function for $G_{\{2\}}$, and use ProbXXY-function for $G_{\{3\}}$

```

In[*]:= Pxyy = FullSimplify[ExpandAll[fG1x*ProbXXY[t1, t2, 4/3]] +
ExpandAll[Integrate[fG1a*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG1c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG1d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1y*ProbXXY[t1, h1, 4/3], {h1, 0, t2 - t1}]] +
ExpandAll[Integrate[fG2a*ProbXXY[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG2b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG2c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG2d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3a*ProbXXY[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3b*ProbXXY[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[
Integrate[fG3c*ProbXXY[h1, h2, 4/3], {h2, 0, t1 - h1}], {h1, 0, t1}]] +
ExpandAll[
Integrate[Integrate[fG3d*ProbXXY[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]]]

```

Out[*]=

$$\begin{aligned}
& \frac{12 (2 + \lambda (10 + 9 \lambda))}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \\
& \frac{3}{32} e^{-\frac{1}{3} (3 t_1 + 2 t_2) (4 + 3 \lambda)} \left(-3 e^{2 t_1 (2 + \lambda)} + \frac{9 e^{\frac{20 t_1}{3} + 2 t_1 \lambda} \lambda}{4 + 3 \lambda} + \frac{15 e^{\frac{8 t_2}{3} + 2 t_2 \lambda} \lambda}{4 + 3 \lambda} + \frac{6 e^{\frac{20 t_1}{3} + 3 t_1 \lambda} \lambda (-2 + 9 \lambda)}{(4 + 3 \lambda)^2} - \right. \\
& \frac{6 e^{\frac{8 t_2}{3} + 2 t_2 \lambda + t_1 (4 + \lambda)} \lambda (-2 + 9 \lambda)}{(4 + 3 \lambda)^2} - \frac{2 e^{t_1 (4 + 3 \lambda)} (-2 + 9 \lambda)}{4 + 3 \lambda} - \\
& \frac{6 e^{\frac{4}{3} (t_1 + 2 t_2) + 2 t_2 \lambda} \lambda}{8 + 9 \lambda} - \frac{3 e^{\frac{2}{3} t_1 (8 + 3 \lambda)} \lambda (-8 + 15 \lambda)}{(4 + 3 \lambda)^2} + \frac{e^{\frac{8 t_1}{3} + 2 t_1 \lambda} (-8 + 15 \lambda)}{4 + 3 \lambda} - \\
& \frac{4 e^{4 t_1 + \frac{8 t_2}{3} + 2 t_2 \lambda} (-8 + \lambda (-29 + 3 \lambda))}{(-8 + 3 \lambda) (4 + 3 \lambda)} - \frac{3 e^{\frac{8 (t_1 + t_2)}{3} + 2 t_2 \lambda} \lambda (-128 + 3 \lambda (400 + 768 \lambda + 81 \lambda^2))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} + \\
& \left. \frac{6 e^{\frac{1}{3} (t_1 + 2 t_2) (4 + 3 \lambda)} (64 + \lambda (-104 + 3 \lambda (-22 + 9 \lambda)))}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} \right)
\end{aligned}$$

$$\begin{aligned}
ln[*] := & \text{LGTProbXYY}[t1_, t2_, \lambda_] := \frac{12 (2 + \lambda (10 + 9 \lambda))}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \\
& \frac{3}{32} e^{-\frac{1}{3} (3 t1 + 2 t2) (4 + 3 \lambda)} \left(-3 e^{2 t1 (2 + \lambda)} + \frac{9 e^{\frac{20 t1}{3} + 2 t1 \lambda} \lambda}{4 + 3 \lambda} + \frac{15 e^{\frac{8 t2}{3} + 2 t2 \lambda} \lambda}{4 + 3 \lambda} + \right. \\
& \frac{6 e^{\frac{20 t1}{3} + 3 t1 \lambda} \lambda (-2 + 9 \lambda)}{(4 + 3 \lambda)^2} - \frac{6 e^{\frac{8 t2}{3} + 2 t2 \lambda + t1 (4 + \lambda)} \lambda (-2 + 9 \lambda)}{(4 + 3 \lambda)^2} - \frac{2 e^{t1 (4 + 3 \lambda)} (-2 + 9 \lambda)}{4 + 3 \lambda} - \\
& \frac{6 e^{\frac{4}{3} (t1 + 2 t2) + 2 t2 \lambda} \lambda}{8 + 9 \lambda} - \frac{3 e^{\frac{2}{3} t1 (8 + 3 \lambda)} \lambda (-8 + 15 \lambda)}{(4 + 3 \lambda)^2} + \frac{e^{\frac{8 t1}{3} + 2 t1 \lambda} (-8 + 15 \lambda)}{4 + 3 \lambda} - \\
& \frac{4 e^{4 t1 + \frac{8 t2}{3} + 2 t2 \lambda} (-8 + \lambda (-29 + 3 \lambda))}{(-8 + 3 \lambda) (4 + 3 \lambda)} - \frac{3 e^{\frac{8 (t1 + t2)}{3} + 2 t2 \lambda} \lambda (-128 + 3 \lambda (400 + 768 \lambda + 81 \lambda^2))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} + \\
& \left. \frac{6 e^{\frac{1}{3} (t1 + 2 t2) (4 + 3 \lambda)} (64 + \lambda (-104 + 3 \lambda (-22 + 9 \lambda)))}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} \right);
\end{aligned}$$

Pxyz

Use ProbXYZ-function for all gene histories

```

In[*]:= Pxyz = FullSimplify[ExpandAll[fG1x*ProbXYZ[t1, t2, 4/3]] +
ExpandAll[Integrate[fG1a*ProbXYZ[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1b*ProbXYZ[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG1c*ProbXYZ[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG1d*ProbXYZ[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG1y*ProbXYZ[t1, h1, 4/3], {h1, 0, t2 - t1}]] +
ExpandAll[Integrate[fG2a*ProbXYZ[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG2b*ProbXYZ[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[Integrate[fG2c*ProbXYZ[h1, h2, 4/3], {h2, 0, t1 - h1}],
{h1, 0, t1}]] + ExpandAll[Integrate[
Integrate[fG2d*ProbXYZ[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3a*ProbXYZ[h1, t1, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[fG3b*ProbXYZ[h1, t2, 4/3], {h1, 0, t1}]] +
ExpandAll[Integrate[
Integrate[fG3c*ProbXYZ[h1, h2, 4/3], {h2, 0, t1 - h1}], {h1, 0, t1}]] +
ExpandAll[
Integrate[Integrate[fG3d*ProbXYZ[h1, h2, 4/3], {h2, 0, t2 - t1}], {h1, 0, t1}]]]

```

Out[*]=

$$\begin{aligned}
& \frac{24 (2 + 3 \lambda)}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \\
& \frac{3}{4} e^{-\frac{1}{3} (3 t_1 + 2 t_2) (4 + 3 \lambda)} \left(e^{2 t_1 (2 + \lambda)} + \frac{24 e^{\frac{20 t_1}{3} + 3 t_1 \lambda} \lambda}{(4 + 3 \lambda)^2} - \frac{24 e^{\frac{8 t_2}{3} + 2 t_2 \lambda + t_1 (4 + \lambda)} \lambda}{(4 + 3 \lambda)^2} + \right. \\
& \frac{3 e^{\frac{2}{3} t_1 (8 + 3 \lambda)} \lambda (-4 + 3 \lambda)}{(4 + 3 \lambda)^2} - \frac{8 e^{t_1 (4 + 3 \lambda)}}{4 + 3 \lambda} + \frac{e^{\frac{8 t_1}{3} + 2 t_1 \lambda} (4 - 3 \lambda)}{4 + 3 \lambda} - \frac{3 e^{\frac{20 t_1}{3} + 2 t_1 \lambda} \lambda}{4 + 3 \lambda} - \\
& \frac{3 e^{\frac{8 t_2}{3} + 2 t_2 \lambda} \lambda}{4 + 3 \lambda} + \frac{8 e^{\frac{1}{3} (t_1 + 2 t_2) (4 + 3 \lambda)} (-2 + 3 \lambda) (8 + 3 \lambda)}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} - \frac{3 e^{4 t_1 + \frac{8 t_2}{3} + 2 t_2 \lambda} \lambda (16 + 3 \lambda)}{(-8 + 3 \lambda) (4 + 3 \lambda)} + \\
& \left. \frac{e^{\frac{4}{3} (t_1 + 2 t_2) + 2 t_2 \lambda} (4 + 9 \lambda)}{8 + 9 \lambda} + \frac{3 e^{\frac{8 (t_1 + t_2)}{3} + 2 t_2 \lambda} \lambda (-64 + 3 \lambda (80 + 3 \lambda (68 + 9 \lambda)))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} \right)
\end{aligned}$$

$$\begin{aligned}
\text{In}[*]:= \text{LGTProbXYZ}[\text{t1_}, \text{t2_}, \lambda_] := & \frac{24 (2 + 3 \lambda)}{(4 + 3 \lambda) (4 + 9 \lambda) (8 + 9 \lambda)} + \\
& \frac{3}{4} e^{-\frac{1}{3} (3 \text{t1} + 2 \text{t2}) (4 + 3 \lambda)} \left(e^{2 \text{t1} (2 + \lambda)} + \frac{24 e^{\frac{20 \text{t1}}{3} + 3 \text{t1} \lambda} \lambda}{(4 + 3 \lambda)^2} - \frac{24 e^{\frac{8 \text{t2}}{3} + 2 \text{t2} \lambda + \text{t1} (4 + \lambda)} \lambda}{(4 + 3 \lambda)^2} + \right. \\
& \frac{3 e^{\frac{2}{3} \text{t1} (8 + 3 \lambda)} \lambda (-4 + 3 \lambda)}{(4 + 3 \lambda)^2} - \frac{8 e^{\text{t1} (4 + 3 \lambda)}}{4 + 3 \lambda} + \frac{e^{\frac{8 \text{t1}}{3} + 2 \text{t1} \lambda} (4 - 3 \lambda)}{4 + 3 \lambda} - \frac{3 e^{\frac{20 \text{t1}}{3} + 2 \text{t1} \lambda} \lambda}{4 + 3 \lambda} - \\
& \frac{3 e^{\frac{8 \text{t2}}{3} + 2 \text{t2} \lambda} \lambda}{4 + 3 \lambda} + \frac{8 e^{\frac{1}{3} (\text{t1} + 2 \text{t2}) (4 + 3 \lambda)} (-2 + 3 \lambda) (8 + 3 \lambda)}{(-8 + 3 \lambda) (-4 + 3 \lambda) (4 + 3 \lambda)} - \frac{3 e^{4 \text{t1} + \frac{8 \text{t2}}{3} + 2 \text{t2} \lambda} \lambda (16 + 3 \lambda)}{(-8 + 3 \lambda) (4 + 3 \lambda)} + \\
& \left. \frac{e^{\frac{4}{3} (\text{t1} + 2 \text{t2}) + 2 \text{t2} \lambda} (4 + 9 \lambda)}{8 + 9 \lambda} + \frac{3 e^{\frac{8 (\text{t1} + \text{t2})}{3} + 2 \text{t2} \lambda} \lambda (-64 + 3 \lambda (80 + 3 \lambda (68 + 9 \lambda)))}{(-4 + 3 \lambda) (4 + 3 \lambda)^2 (4 + 9 \lambda)} \right);
\end{aligned}$$

Generic local identifiability of model parameters from JC69 site pattern probabilities

Vector of site pattern probabilities as a function of t1, t2, and λ

```

In[*]:= f = {LGTProbXXX[t1, t2, λ], LGTProbXXY[t1, t2, λ],
             LGTProbXYY[t1, t2, λ], LGTProbXYZ[t1, t2, λ]};

```

Jacobian matrix (4 x 3 matrix)

```

In[*]:= J = D[f, {{t1, t2, λ}}];

```

Generic rank of Jacobian matrix

```

In[*]:= MatrixRank[J]
Out[*]=

```

3

Rank of Jacobian matrix at a generic admissible point

```

In[*]:= MatrixRank[J /. {t1 -> 1, t2 -> 2, λ -> 0.5}]
Out[*]=

```

3